

## SAFETY DATA SHEET

Revision Date 24-Dec-2021

Revision Number 4

### 1. Identification

**Product Name** Ethanol-d

**Cat No. :** AC166320000; AC166320250; AC166321000

**Synonyms** Ethyl alcohol-d

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable liquids

Category 2

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor

**Precautionary Statements****Prevention**

Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Wear protective gloves/protective clothing/eye protection/face protection

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component        | CAS No   | Weight % |
|------------------|----------|----------|
| Ethyl [2]alcohol | 925-93-9 | 100      |

### 4. First-aid measures

|                                            |                                                                                                                                              |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                        |
| <b>Inhalation</b>                          | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Get medical attention.                   |
| <b>Ingestion</b>                           | Clean mouth with water. Get medical attention.                                                                                               |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                        |

### 5. Fire-fighting measures

|                                       |                                                                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                                                                       |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Flash Point</b>                      | 12 °C / 53.6 °F          |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | 363 °C / 685.4 °F        |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | 19%                      |
| <b>Lower</b>                            | 3.3%                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Flammable. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
0

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b> | See Section 12 for additional Ecological Information.                                  |

|                                             |                                                                                                                                                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Take precautionary measures against static discharges. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Flammables area. Store under an inert atmosphere. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Acids. Ammonia. Peroxides. Acid anhydrides. Acid chlorides. Metals.      |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                                                                  |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                                                                    |
| <b>Engineering Measures</b>          | Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. |
| <b>Personal Protective Equipment</b> |                                                                                                                                                                                                                  |
| <b>Eye/face Protection</b>           | Wear appropriate protective eyeglasses or chemical safety goggles as described by                                                                                                                                |

|                                 |                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------|
|                                 | OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                 |
| <b>Respiratory Protection</b>   | No protective equipment is needed under normal use conditions.                            |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                    |

## 9. Physical and chemical properties

|                                               |                             |
|-----------------------------------------------|-----------------------------|
| <b>Physical State</b>                         | Liquid                      |
| <b>Appearance</b>                             | Colorless                   |
| <b>Odor</b>                                   | No information available    |
| <b>Odor Threshold</b>                         | No information available    |
| <b>pH</b>                                     | No information available    |
| <b>Melting Point/Range</b>                    | No data available           |
| <b>Boiling Point/Range</b>                    | 78 °C / 172.4 °F @ 760 mmHg |
| <b>Flash Point</b>                            | 12 °C / 53.6 °F             |
| <b>Evaporation Rate</b>                       | No information available    |
| <b>Flammability (solid,gas)</b>               | Not applicable              |
| <b>Flammability or explosive limits</b>       |                             |
| Upper                                         | 19%                         |
| Lower                                         | 3.3%                        |
| <b>Vapor Pressure</b>                         | No information available    |
| <b>Vapor Density</b>                          | 1.62                        |
| <b>Specific Gravity</b>                       | 0.800                       |
| <b>Solubility</b>                             | No information available    |
| <b>Partition coefficient; n-octanol/water</b> | No data available           |
| <b>Autoignition Temperature</b>               | 363 °C / 685.4 °F           |
| <b>Decomposition Temperature</b>              | No information available    |
| <b>Viscosity</b>                              | No information available    |
| <b>Molecular Formula</b>                      | C2 H5 D O                   |
| <b>Molecular Weight</b>                       | 47.07                       |

## 10. Stability and reactivity

|                                         |                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                        |
| <b>Stability</b>                        | Stable under normal conditions. Hygroscopic.                                                      |
| <b>Conditions to Avoid</b>              | Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Acids, Ammonia, Peroxides, Acid anhydrides, Acid chlorides, Metals                                |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                           |
| <b>Hazardous Polymerization</b>         | No information available.                                                                         |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                     |

## 11. Toxicological information

### Acute Toxicity

|                                                                                                          |                                                             |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Product Information</b>                                                                               | No acute toxicity information is available for this product |
| <b>Component Information</b>                                                                             |                                                             |
| <b>Toxicologically Synergistic Products</b>                                                              | No information available                                    |
| <b><u>Delayed and immediate effects as well as chronic effects from short and long-term exposure</u></b> |                                                             |

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component        | CAS No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|----------|------------|------------|------------|------------|------------|
| Ethyl [2]alcohol | 925-93-9 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                                                                            |
| <b>Reproductive Effects</b>                       | No information available.                                                                                           |
| <b>Developmental Effects</b>                      | No information available.                                                                                           |
| <b>Teratogenicity</b>                             | No information available.                                                                                           |
| <b>STOT - single exposure</b>                     | None known                                                                                                          |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                          |
| <b>Aspiration hazard</b>                          | No information available                                                                                            |
| <b>Symptoms / effects, both acute and delayed</b> | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                            |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                      |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                                                                 |
|--------------------------------------|-----------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely based on information available.         |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                       |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its volatility. |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|               |        |
|---------------|--------|
| UN-No         | UN1170 |
| Hazard Class  | 3      |
| Packing Group | II     |

### TDG

|               |        |
|---------------|--------|
| UN-No         | UN1170 |
| Hazard Class  | 3      |
| Packing Group | II     |

### IATA

|                      |         |
|----------------------|---------|
| UN-No                | UN1170  |
| Proper Shipping Name | ETHANOL |
| Hazard Class         | 3       |
| Packing Group        | II      |

### IMDG/IMO

|                      |         |
|----------------------|---------|
| UN-No                | UN1170  |
| Proper Shipping Name | ETHANOL |
| Hazard Class         | 3       |
| Packing Group        | II      |

## 15. Regulatory information

### United States of America Inventory

| Component        | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|------------------|----------|------|-----------------------------------------------|-----------------------------|
| Ethyl [2]alcohol | 925-93-9 | -    | -                                             | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component        | CAS No   | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|------------------|----------|-----|------|-----------|-------|------|------|------|-------|------|
| Ethyl [2]alcohol | 925-93-9 | -   | -    | 213-128-9 | -     | -    |      | -    | -     | -    |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA** - Occupational Safety and Health Administration Not applicable

**CERCLA** Not applicable

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations** Not applicable

### U.S. Department of Transportation

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

Mexico - Grade No information available

Authorisation/Restrictions according to EU REACH

Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component        | CAS No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Ethyl [2]alcohol | 925-93-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component        | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Ethyl [2]alcohol | 925-93-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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Email: EMSDS.RA@thermofisher.com

**Revision Date** 24-Dec-2021

**Print Date** 24-Dec-2021

**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**